BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Objective Type

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. If y = tanh^-1 x, then dy/dx =:

(A) 1 / (1 + x^2)
(B) 1 / (1 - x^2)
(C) -1 / (1 + x^2)
(D) -1 / (1 - x^2)

Q2. Integral from 0 to 3 of dx / (x^2 + 9) =:

(A) pi / 4
(B) -pi / 4
(C) 0
(D) pi / 12

Q3. Integral of e^x (sin x + cos x) dx =:

(A) e^x cos x + c
(B) e^x sin x
(C) e^x sin x + c
(D) e^x cos x

Q4. d/dx (sin sqrt(x)) =:

(A) cos sqrt(x)
(B) cos sqrt(x) . 1 / sqrt(x)
(C) sqrt(x) cos sqrt(x)
(D) cos sqrt(x) . 1 / (2 sqrt(x))

Q5. d/dx (a^(lambda x)) =:

(A) a^(lambda x)
(B) a^(lambda x) lambda ln a
(C) lambda a^(lambda x) ln a
(D) a^(lambda x) / (lambda ln a)

Q6. The perimeter P of a square as a function of its area A is given as:

(A) 4A
(B) 4 sqrt(A)
(C) 2A
(D) 2 sqrt(A)

Q7. If y = x^2 - 1, then dy =:

(A) xdx + c
(B) (x - 1)dx
(C) 2x dx + c
(D) 2x dx

Q8. Integral of 2 / (x + 2) dx =:

(A) ln |x + 2| + c
(B) ln |x + 2|^2 + c
(C) 1 / ln |x + 2|
(D) 2 ln x + c

Q9. Integral of 1 / cos^2 x dx =:

(A) 1 / sin^2 x + c
(B) tan x + c
(C) sec^2 x + c
(D) cosec^2 x + c

Q10. Domain of cosine function y = cos x is:

(A) Real numbers
(B) [-1, 1]
(C) (0, infinity)
(D)]-1, 1[

Q11. Two lines a1x + b1y + c1 = 0 and a2x + b2y + c2 = 0 are parallel if:

(A) a1a2 + b1b2 = 0
(B) a1a2 - b1b2 = 0
(C) a1b2 - a2b1 = 0
(D) a1b2 + a2b1 = 0

Q12. For parabola, value of eccentricity e is:

(A) e = 0
(B) e < 1
(C) e > 1
(D) e = 1

Q13. A region is said to feasible region which is restricted to:

(A) I quadrant
(B) II quadrant
(C) III quadrant
(D) IV quadrant

Q14. Focus of the parabola x^2 - 5y = 0 is:

(A) (5/4, 0)
(B) (0, 5/4)
(C) (0, -5/4)
(D) (-5/4, 0)

Q15. The lines represented by ax^2 + 2hxy + by^2 = 0 are imaginary if :

(A) h^2 - ab = 0
(B) h^2 - ab < 0
(C) h^2 - ab > 0
(D) h^2 - ab != 0

Q16. A vector perpendicular to both vectors a and b is:

(A) a . b
(B) a x b
(C) ab / |a|
(D) b . a

Q17. Inclination of the line joining the points (4, 6) and (4, 8) is:

(A) 90 degrees
(B) 45 degrees
(C) 30 degrees
(D) Undefined

Q18. The coordinate of the vertices of the hyperbola x^2 / a^2 - y^2 / b^2 = 1 is:

(A) (0, +-b)
(B) (+-b, 0)
(C) (0, +-a)
(D) (+-a, 0)

Q19. If u, v and w are coterminous edges of a tetrahedron, then its volume is:

(A) [u v w]
(B) 1/3 [u v w]
(C) 1/6 [u v w]
(D) 1/9 [u v w]

Q20. An angle in a semicircle is of measure:

(A) 90 degrees
(B) 60 degrees
(C) 45 degrees
(D) 30 degrees

Essay Type

Time Allowed: 2.30 Hrs. | **Maximum Marks:** 80

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT questions:

(i) For f(x) = (2x+1)/(x-1), find f^-1(x)

(ii) Evaluate Lim theta -> 0 (1 - cos theta) / sin theta

(iii) Discuss the continuity of f(x) at x = c = 2, f(x) = { 2x+5 if x <= 2; 4x+1 if x > 2 }

(iv) Differentiate w.r.t 'x' (x-5)(3-x)

(v) Find dy/dx if y^2 - xy - x^2 + 4 = 0

(vi) Differentiate w.r.t 'theta' (sin 2theta - cos 3theta)^2

(vii) Find dy/dx if y = x^2 ln 1/x

(viii) Find dy/dx if y = (2x+5)^(3/2)

(ix) Apply Maclaurin series expansion to prove that cos x = 1 - x^2/2! + x^4/4! - x^6/6! + - - - - - -

(x) Find extreme values for f(x) = x^2 - x - 2

(xi) Define feasible region.

(xii) Graph the inequality x + 2y <= 6

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Find delta y and dy in the case y = x^2+2x when x changes from 2 to 1.8

(ii) Evaluate integral of 1 / (sqrt(x)(sqrt(x) + 1)) dx, x > 0

(iii) Evaluate integral of a^(x^2) x dx (a > 0, a != 1)

(iv) Evaluate integral of sqrt(4 - 5x^2) dx

(v) Evaluate integral from 0 to pi/6 of x cos x dx

(vi) Find area below the curve y = 3 sqrt(x) and above the x-axis between x = 1 and x = 4

(vii) Solve the differential equation x^2 (2y + 1) dy/dx - 1 = 0

(viii) Find the position vector of the point of division of the line segments joining the following pair of points, in the given ratio, point C with position vector 2i - 3j and point D with position vector 3i + 2j in the ratio 4 : 3.

(ix) If u = 2i + 3j + 4k, v = -i + 3j - k and w = i + 6j + z k represent the sides of a triangle, find the value of z.

(x) Find the angle between the vectors u = 2i - j + k and v = -i + j.

(xi) If a = 4i + 3j + k and b = 2i - j + 2k, find a unit vector perpendicular to both a and b. Also find the sine of angle between the vectors a and b.

(xii) Find the area of the triangle with vertices A (1, -1, 1), B (2, 1, -1) and C (-1, 1, 2)

Q4. (Marks 18) Write short answers to any NINE (9) questions:

(i) Show that the points A (0, 2), B (sqrt(3), -1) and C (0, -2) are vertices of a right triangle. D (-6, 4) are parallel.

(ii) Find k so that the line joining A (7, 3), B (k, -6) and line joining C (-4, 5), D (-6, 4) are parallel.

(iii) Find an equation of line if its slope is 2 and y-intercept is 5.

(iv) Transform the equation 5x - 12y + 39 = 0 into two-intercept form.

(v) Find the distance from the points P (6, -1) to the line 6x - 4y + 9 = 0.

(vi) Find the point of intersection of lines 3x + y + 12 = 0 and x + 2y - 1 = 0.

(vii) Find the angle between the lines represented by x^2 - xy - 6y^2 = 0.

(viii) Find an equation of circle with centre at (sqrt(2), 3 sqrt(3)) and radius 2 sqrt(2)

(ix) Find centre and radius of circle x^2 + y^2 + 12x - 10y = 0

(x) Find vertex and directrix of parabola x^2 = 16y

(xi) Find the focus and vertex of parabola x^2 = 4(y - 1)

(xii) Find centre and foci of 4x^2 + 9y^2 = 36

(xiii) Find eccentricity and vertices of y^2/16 - x^2/9 = 1

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [5] Evaluate Lim theta -> 0 (1 - cos p theta) / (1 - cos q theta)

(b) [5] If y/x = tan^-1 x/y then prove that dy/dx = y/x

Q6.

(a) [5] Evaluate integral of x / (x^4 + 2x^2 + 5) dx

(b) [5] Find equations of two parallel lines perpendicular to 2x - y + 3 = 0 such that the product of the x-intercept and y-intercept of each is 3.

Q7.

(a) [5] Evaluate integral from 0 to pi/4 of (1 + cos^2 theta) tan^2 theta d theta

(b) [5] Minimize z = 2x + y subject to the constraints x + y >= 3, 7x + 5y <= 35, x >= 0, y >= 0.

Q8.

(a) [5] If y = (cos^-1 x)^2, prove that (1 - x^2) y2 - x y1 - 2 = 0

(b) [5] Find equations of the tangents to the circle x^2 + y^2 = 2 and parallel to the line x - 2y + 1 = 0

Q9.

(a) [5] Find the volume of the tetrahedron with the vertices (0, 1, 2), (3, 2, 1), (1, 2, 1) and (5, 5, 6)

(b) [5] Find the centre, foci, eccentricity and directrices of ellipse (2x-1)^2/4 + (y+2)^2/16 = 1